

# PEAD Strategies Overview

One at-a-glance catalog of every strategy design in this project: what it does, how it performed, and whether it needs a GPU.

Companion to *PEAD\_Strategy\_Showcase.pdf*, *PEAD\_Strategy\_Development.pdf*, and *PEAD\_Options\_Strategy\_Report.pdf* -- this report is the index, not the deep-dive.

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This project has built **9 equity strategy designs** (Strategies 1-8, with Strategy 6 also having a beta-overlay variant), an **aggressive, higher-return/lower-Sharpe variant of each of those 8** (same signal and construction, bigger position size and leverage cap), and explored **one options strategy across 4 GPU-tiered stages** -- Tier 1 through Tier 4 are not 4 separate, independent options strategies; they are one lineage, each tier built only because the tier before it left a concrete question unanswered. Of those 5 options rows, only Tiers 1, 1.5, and 4 currently have a complete, real backtested Sharpe/return/drawdown; Tiers 2 and 3 are still parameter-search/screening stages -- their status column says so explicitly rather than showing a number that doesn't exist yet.

**How to read "GPU needed?"** Every equity strategy 1-7 and options Tier 1/1.5 runs fine on a laptop CPU in seconds. GPU only matters for the compute-heavy tiers built for the dedicated 5090 machine: Strategy 8's walk-forward model training and parameter sweep, and options Tiers 2-4's large batched tensor searches.

## Equity strategies (9 designs)

| Strategy                             | What it does                                                                                                                                                                                      | Ann. Return | Ann. Vol | Sharpe | Max DD  | GPU?                                                                                                          |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|--------|---------|---------------------------------------------------------------------------------------------------------------|
| <b>1. Extreme Decile L/S</b>         | Long the most positive earnings-surprise decile, short the most negative, holding each decile/size/book-to-market cell for its own empirically-best number of days.                               | 4.91%       | 5.32%    | 0.93   | -18.48% | No (CPU, seconds)                                                                                             |
| <b>2. Rank-Weighted</b>              | Trades every SUE decile at once, weighted continuously by rank distance from the median for broader diversification.                                                                              | 3.88%       | 3.86%    | 1.01   | -13.96% | No (CPU, seconds)                                                                                             |
| <b>3. Balanced</b>                   | The extreme-decile trade, neutralized by size quintile each quarter so no single size bucket dominates the book.                                                                                  | 4.74%       | 5.26%    | 0.91   | -17.53% | No (CPU, seconds)                                                                                             |
| <b>4. Trimmed/Tilted</b>             | Drops the low-conviction middle of the SUE distribution, tilts weight toward the extremes, and trims lowest-conviction positions first when a cap binds.                                          | 5.26%       | 4.41%    | 1.18   | -13.07% | No (CPU, seconds)                                                                                             |
| <b>5. + Sector-Neutral</b>           | Strategy 4 plus Fama-French 12-sector neutrality, fixing a hidden sector-concentration risk -- best Sharpe and shallowest drawdown of the hand-built designs.                                     | 4.06%       | 3.02%    | 1.33   | -10.83% | No (CPU, seconds)                                                                                             |
| <b>6. + Leverage Cap 2.5x</b>        | Strategy 5 with its leverage cap raised from 1.5x to 2.5x, since diagnostics showed nothing left for the tighter cap to protect against.                                                          | 6.49%       | 4.94%    | 1.30   | -16.63% | No (CPU, seconds)                                                                                             |
| <b>6+Beta. Beta Overlay</b>          | Strategy 6's untouched alpha book plus a separate 0.5x-NAV market-index position held alongside it, pushing return toward equity-like levels.                                                     | 10.08%      | 10.51%   | 0.97   | -23.23% | No (CPU, seconds)                                                                                             |
| <b>7. Unconstrained Net Exposure</b> | Strategy 5/6's signal without forcing long and short dollar totals to match each quarter -- a documented cautionary result, not a recommendation.                                                 | 7.30%       | 6.56%    | 1.11   | -15.54% | No (CPU, seconds)                                                                                             |
| <b>8. GPU/ML-Tiered</b>              | Blends SUE rank with a walk-forward-trained neural signal built on trailing momentum/volatility features, with sizing and leverage chosen by a real grid search instead of hand-picked constants. | 3.51%       | 3.77%    | 0.93   | -12.93% | Yes (walk-forward training + parameter sweep are GPU-batched -- 12.3x faster on a Colab T4 than this dev CPU) |

Table 1. Figures 1996-2013, market-adjusted (raw for Strategy 7) daily returns, trailing-NAV compounding, \$10,000,000 starting capital. Recommended for live consideration: Strategy 6 (pure alpha) or Strategy 6+Beta (higher return, more market-correlated risk).

**Strategy 8 status:** Real run, full sample: the GPU param sweep itself picked `ml_blend_weight=0.0` (pure SUE rank beat every ML-blended variant) and holdout Sharpe (0.42) trailed search-window Sharpe (1.14) -- the ML signal does not add real value in this data and Strategy 8 underperforms Strategies 5/6. Kept as a documented result, like the EAR and SUE-reversal dead ends. Walk-forward mean out-of-sample IC across all 15 freshly-trained yearly folds was 0.0095 -- essentially no signal. Confirmed on a real Colab T4 GPU as well as this CPU dev machine (identical numbers on both, 12.3x faster on the T4: 660s -> 55s for the walk-forward training step alone).

### Strategy Comparison: Return, Risk-Adjusted Return, and Drawdown

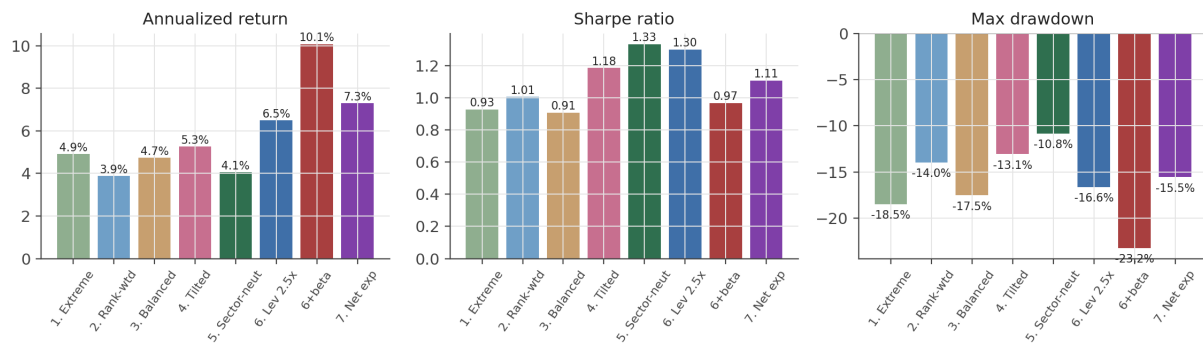


Figure 1. Return/Sharpe/drawdown comparison across all 8 hand-built equity strategies.

## Aggressive equity variants (same 8 designs, sized for more return)

Each of the 8 hand-built designs above also has an **aggressive variant**: identical signal, trim, tilt, and neutralization -- only position size (base\_unit\_fraction) and the gross-leverage cap are pushed harder, real-backtested on the same data. Selected to maximize return subject to Sharpe  $\geq 0.6$  AND max drawdown  $\geq -40\%$  (the drawdown floor binds first for every variant except the beta overlay, which is why none of these Sharpe ratios are actually as low as 0.6). See *PEAD\_Strategy\_Showcase.pdf*'s "Aggressive variants" section for full methodology, the sizing-sweep chart, and caveats (higher liquidity-cap bite, deeper drawdowns). Strategy 8 (GPU/ML-tiered) is not included -- its own hyperparameter search is a different kind of lever than base\_unit\_fraction/leverage and is out of scope for this pass.

| Strategy                           | Ann. Return | Ann. Vol | Sharpe | Max DD  |
|------------------------------------|-------------|----------|--------|---------|
| 1. Extreme Decile L/S -- Aggr.     | 9.53%       | 12.24%   | 0.80   | -36.20% |
| 2. Rank-Weighted -- Aggr.          | 10.84%      | 12.40%   | 0.89   | -38.76% |
| 3. Balanced -- Aggr.               | 8.45%       | 12.26%   | 0.72   | -37.45% |
| 4. Trimmed/Tilted -- Aggr.         | 10.70%      | 12.25%   | 0.89   | -40.00% |
| 5. + Sector-Neutral -- Aggr.       | 12.15%      | 12.20%   | 1.00   | -36.46% |
| 6. + Leverage Cap -- Aggr.         | 12.15%      | 12.20%   | 1.00   | -36.46% |
| 6+Beta. Beta Overlay -- Aggr.      | 15.01%      | 13.70%   | 1.09   | -36.98% |
| 7. Unconstrained Net Exp. -- Aggr. | 11.78%      | 11.85%   | 1.00   | -32.97% |

Table 1b. Same figures/assumptions as Table 1, \$10,000,000 starting capital, base\_unit\_fraction and leverage cap searched up to a 6x realistic institutional ceiling. Compare row-by-row against Table 1 above: every aggressive variant roughly doubles or more its own baseline's return.

Baseline vs. Aggressive Variant, All 8 Strategies

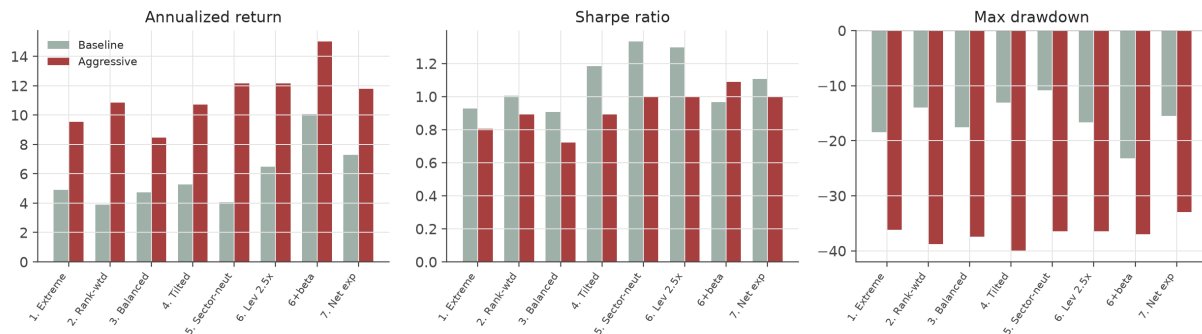


Figure 1b. Baseline vs. aggressive, all 8 designs, side by side.

## Options strategy (1 lineage, 5 tiers)

| Tier                                                 | What it does                                                                                                                                                             | Ann. Return | Ann. Vol | Sharpe | Max DD  | GPU?                                                     |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|--------|---------|----------------------------------------------------------|
| <b>Tier 1. Near-ATM Heuristic</b>                    | Buys a near-50-delta call (long-SUE deciles) or put (short-SUE deciles) at earnings and holds to a fixed horizon, with real capital sizing and trading costs.            | 9.51%       | 9.16%    | 1.04   | -25.90% | No (CPU, seconds)                                        |
| <b>Tier 1.5. Kelly-Optimal Selector</b>              | Scores every strike in the real day-0 chain by expected log-growth against an empirical, walk-forward return distribution, sized at half-Kelly.                          | 15.31%      | 4.03%    | 3.56   | -21.36% | Optional (ran on CPU; GPU-batchable at full-chain scale) |
| <b>Tier 2. Dynamic Exit Optimizer</b>                | Searches for the best stop-loss/profit-target exit rule per decile bucket instead of a fixed holding horizon.                                                            | N/A         | N/A      | N/A    | N/A     | Yes (dense position x day x parameter-combo tensor scan) |
| <b>Tier 3. Sizing Sweep + ML Selector</b>            | A GPU grid search over trim/tilt/horizon to screen sizing constants, plus a planned Optuna/PyTorch model testing IV/liquidity/sector features against decile membership. | N/A         | N/A      | N/A    | N/A     | Yes (GPU-batched sweep; Optuna + PyTorch training)       |
| <b>Tier 4. Joint Equity+Options +Beta Allocation</b> | Searches the best combined weighting across the equity alpha book, the options sleeve, and the market-beta overlay held together.                                        | 14.56%      | 7.64%    | 1.82   | -16.63% | Yes (largest tensor sweep in the project)                |

Table 2. Figures at the 60-day horizon, 1996-2013 ex-2011, real capital sizing and trading costs, \$10,000,000 starting capital. Tier 1.5 is the currently-recommended options design.

**Tier 2. Dynamic Exit Optimizer status:** Parameter search only, run at small/mock scale so far -- not yet folded into a full capital-sized backtest.

**Tier 3. Sizing Sweep + ML Selector status:** Sizing sweep completed on real full-scale data as a screening pass (not a final Sharpe); the ML selector has not been run.

**Tier 4 leverage caveat:** the best combo found ( $w_{\text{equity}}=2.0$ ,  $w_{\text{options}}=1.0$ ,  $w_{\text{beta}}=0.25$ ) assumes weights above 1.0x are literally free, unlimited leverage on top of Strategy 6's already-2.5x internal leverage. Treat the 1.82 Sharpe / 14.56% return as an upper-bound research signal about where blending helps, not an achievable number, until a real cost-of-leverage term is added.

### Tier 1 vs. Tier 1.5, by hold horizon

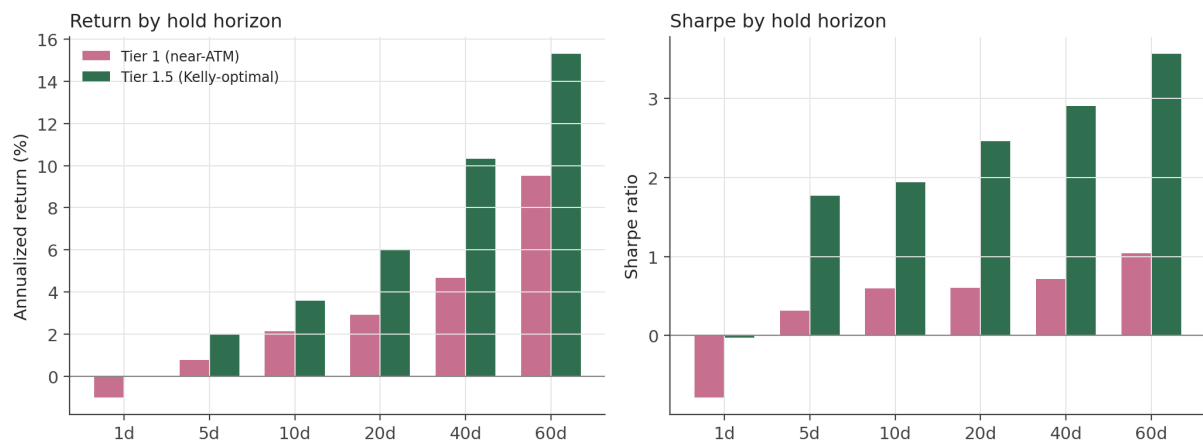


Figure 2. Tier 1 (near-ATM heuristic) vs. Tier 1.5 (Kelly-optimal) by holding horizon.